

Modelling the Solar System

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ABSTRACT

This project explores the behaviour of celestial bodies in orbit.

Keywords: Orbit, Integration, Energy

1 INTRODUCTION

This project explores the behaviour of celestial bodies orbiting a central body, in this case, this is the solar system. There are three experiments performed, exploring the accuracy of the simulation in determining orbital periods, conserving energy, and modelling a probe on a low energy, Hohmann trajectory from the orbit of the Earth to that of Mars. Each experiment aims to provide clear data that can be compared to real world findings in order to assess the suitability of the simulation in modelling the solar system.

2 METHODS

In the main source code, there are two classes - Body and Simulation - representing a single celestial body and the n-body simulation respectively.

In the Body class, there are methods to update the motion of a body using any of three integration methods: Beeman, Euler-Cromer, and Direct Euler.

The Beeman method calculates the position using both the current and previous accelerations³. The new position is then used to calculate the new acceleration, and both are used to calculate the new velocity.

$$\vec{r}(t + \Delta t) = \vec{r}(t) + \vec{v}(t)\Delta t + \frac{1}{6}[4\vec{a}(t) - \vec{a}(t - \Delta t)]\Delta t^2 \quad (1)$$

$$\vec{v}(t + \Delta t) = \vec{v}(t) + \frac{1}{6}[2\vec{a}(t) + \vec{a}(t) - \vec{a}(t - \Delta t)]\Delta t \quad (2)$$

The Euler-Cromer method works by updating the velocity at each time step and then using the new velocity to update the position.

$$\vec{v}(t + \Delta t) = \vec{v}(t) + \vec{a}(t)\Delta t \quad (3)$$

$$\vec{r}(t + \Delta t) = \vec{r}(t) + \vec{v}(t + \Delta t)\Delta t \quad (4)$$

The Direct Euler method updates the position using the previous velocity and then adjusts the velocity. The key difference between Euler-Cromer and Direct Euler is the order in which position and velocity are updated.

$$\vec{r}(t + \Delta t) = \vec{r}(t) + \vec{v}(t)\Delta t \quad (5)$$

$$\vec{v}(t + \Delta t) = \vec{v}(t) + \vec{a}(t)\Delta t \quad (6)$$

This class allows you to 'check orbital period'. If the body is not the Sun or a satellite, and the y-component of the position vector has crossed from a negative value to a positive one whilst being on the positive x axis, the system will recognise this as a full orbit and store the period as the elapsed time.

In the Simulation class, the method 'add satellite' is specifically for use in experiment 3 in order to allow the experiment's code to be concise. The 'calc acceleration' method iterates over all bodies other than the one in question to find the vector between the two bodies and sums the individual contributions to the acceleration of that body.

$$\vec{a}_i = \sum_{\substack{j=1 \\ j \neq i}}^N Gm_j \frac{\vec{r}_j - \vec{r}_i}{\|\vec{r}_j - \vec{r}_i\|^3} \quad (7)$$

The 'calc pe' method sums the Gravitational potential energy of each body in the system.

$$U_{ij} = -\frac{Gm_i m_j}{r_{ij}} \quad (8)$$

where $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$ is the distance between bodies i and j . In the code, this is implemented as a double sum in order to avoid double-counting of any pair of bodies in reverse order.

$$U = -\sum_{i=1}^N \sum_{j=i+1}^N \frac{Gm_i m_j}{|\mathbf{r}_j - \mathbf{r}_i|} \quad (9)$$

The simulation is able to run either with animation, or without - which becomes useful for the experiments as they only need the data output. The 'run animated' method is similar to the 'run simulation' method with the added functionality of producing an animated plot of the planets orbiting the Sun as coloured patches with trail lines showing their path.

The 'plot energy' method will only work if the simulation has stepped forward as, if the time history is blank, there will be no data to plot. It will plot the total energy of the system against the time passed.

A default JSON file has been provided in case any input file is not accessible, which will be created by the 'ensure JSON' method if the 'parameters solar' file is not accessible.

2.1 Experiment 1: Orbital Periods

This experiment runs the Beeman simulation and compares the detected orbital periods against NASA's published values⁴. The time step is 600 steps per Earth year to produce accurate results without excessive runtime. The simulation runs for 13 years so that at least one orbit can be completed by Jupiter.

The energy write interval is 100 rather than 10 to obtain more data over the same amount of time, allowing for more accurate orbital periods for each body.

A bar chart is generated to visually compare the data, showing the NASA and the simulated data side by side for each planet in the simulation.

2.2 Experiment 2: Energy Conservation and Alternative Integration Methods

The simulation runs three times using different integration methods and compares how well each approach conserves total energy over time, measuring how effective they are in accurately modelling orbital motion. This simulation will run for 20 years allowing for the energy drift in Direct Euler integration to be observed properly.

All energies are normalised using

$$\Delta E_{\text{norm}}(t) = \frac{E(t) - E_0}{|E_0|} \quad (10)$$

where $E(t)$ is the total energy at time t and E_0 is the initial total energy at $t = 0$.

A plot is then made to show the normalised fractional data from all three methods on the same axes as a means of comparison.

A second plot shows the Beeman method's data in more detail, zoomed in to reveal fine-scale oscillations in the total energy output.

2.3 Experiment 3: Satellite to Mars

This models a small probe launched from just outside of Earth and searches for an initial velocity that will bring it close to Mars. The Hohmann transfer orbit is the most fuel-efficient path between two circular orbits² and so has been chosen for this experiment. The satellite is launched tangentially relative to Earth. Earth travels at its circular orbit speed, which can be obtained using Kepler's third law:

$$v_{\oplus} = \sqrt{\frac{G}{r_{\oplus}}} = \sqrt{\frac{4\pi^2}{1.0}} \approx 6.28 \text{ AU yr}^{-1} \quad (11)$$

If the probe were launched at this speed it would follow the Earth's orbit and never reach Mars, hence why a prograde burn is applied⁶. This can be seen as a range of speed multipliers, called boost factors in the code, which are tested to find when the satellite is closest to Mars.

The probe's launch velocity therefore becomes:

$$v_{\text{launch}} = f \times v_{\oplus}, \quad f > 1 \quad (12)$$

where f represents the boost factor.

The satellite is launched from just outside of Earth's surface, this small radial offset of $y = +0.001$ AU aims to avoid division by zero.

There are two new functions in this code. The first computes Earth's orbital speed. The second is a function that searches the list of bodies and returns the object whose name attribute matches a given string. This becomes useful as, after the body list is populated, there is no direct way to retrieve a specific body by name without iterating through it and so this function is called repeatedly during the code to locate the Mars and probe objects at each time step.

The Hohmann transfer optimal speed ratio is approximately:

$$\frac{v_{\text{launch}}}{v_{\oplus}} = \sqrt{\frac{2r_{\text{M}}}{r_{\oplus} + r_{\text{M}}}} \approx 1.099 \quad (13)$$

The 'boost factors' that will be tested as the speed multipliers range from 1.05 to 1.20, in order to provide 16 data points around the estimate of 1.099 while being a wide enough range to spot the true optimal velocity. This is because the Hohmann transfer assumes perfectly circular orbits and the planets starting in exactly the right relative positions, whereas the simulation models all of the planets starting on the x-axis simultaneously, which is unrealistic.

For each factor, a fresh simulation is constructed and run without animation for up to 3 years. At each time step, the code will record the distance between the probe and Mars with the aim of storing the closest approach distance and the time at which it occurs for each factor. The distance between the two bodies is calculated.

$$d(t) = |\mathbf{r}_{\text{probe}}(t) - \mathbf{r}_{\text{Mars}}(t)| \quad (14)$$

The decision to run each trial independently is to avoid any chance of not using the same starting configuration each time. Once the most suitable boost factor is identified, the simulation runs once more using that factor, saving the full position history of every body. 'this approach avoids using unnecessary memory, allowing the code to run faster than if the history were saved for every boost factor.

The simulation uses a smaller time step than previous experiments for more accurate trajectories near the closest approach, as this is where small differences in position will matter the most. It uses a simulation time of 3 years: the orbital period of Mars is 1.88 years, if the probe has not reached Mars within this time period, it is either in the wrong orbit or has passed Mars. The actual travel time of a launch to Mars using the Hohmann transfer is around 8.5 months¹, giving an idea of what to expect.

Also, the probe is modelled with a negligible mass of 10^{-25} - a realistic spacecraft mass in solar masses is so small that its gravitational effect on the planets would be physically meaningless.

RESULTS

The main 'solar simulation' code produces an animated view of the solar system orbiting the Sun, assuming that all planets begin in alignment. Figure 1 shows the planets at an arbitrary time of $t = 8.70$ years. Although the planets begin aligned, this changes as the simulation runs due to their differing orbital periods. Figure 2 depicts the total energy of the system over time - the maximum energy level is reached

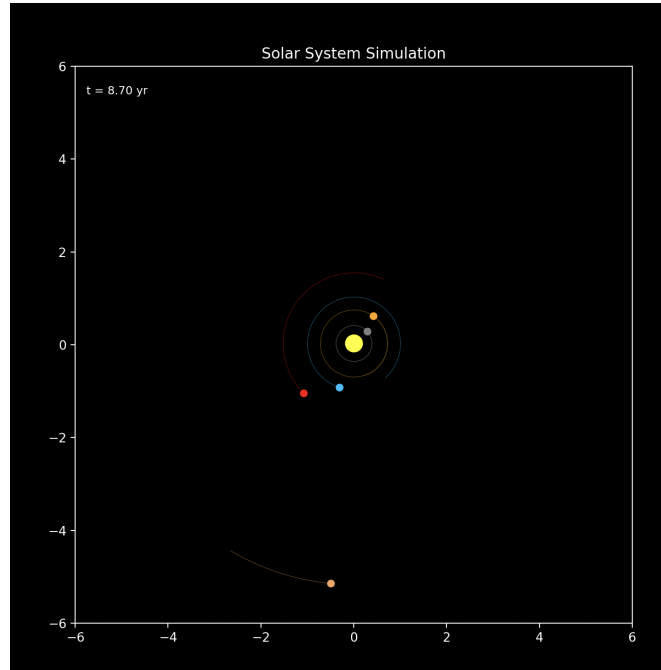


Figure 1. A frame of the solar system simulation.

within ~ 0.1 years or ~ 60 iterations of the system. It's important to note that the total energy is negative due to the convention of zero potential energy occurring at infinite distance from an object.

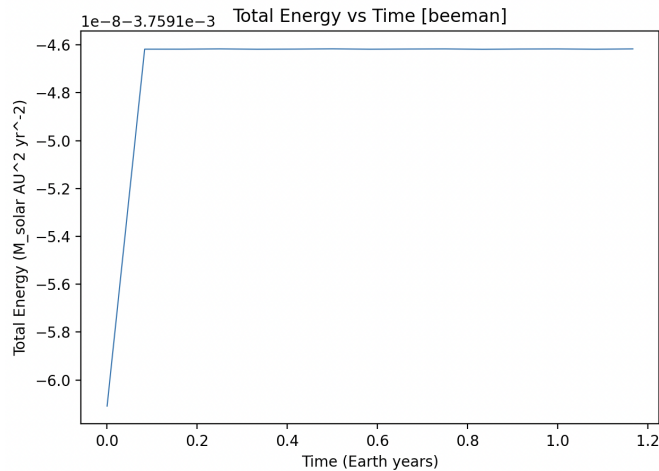


Figure 2. Total energy of system using Beeman method against time.

2.4 Experiment 1

A bar chart was created to compare the data produced by the simulation with NASA's published data. Figure 3 shows that NASA's data for orbital periods is slightly larger than the simulation's. This discrepancy appears to increase as the total orbital period increases, albeit remaining small.

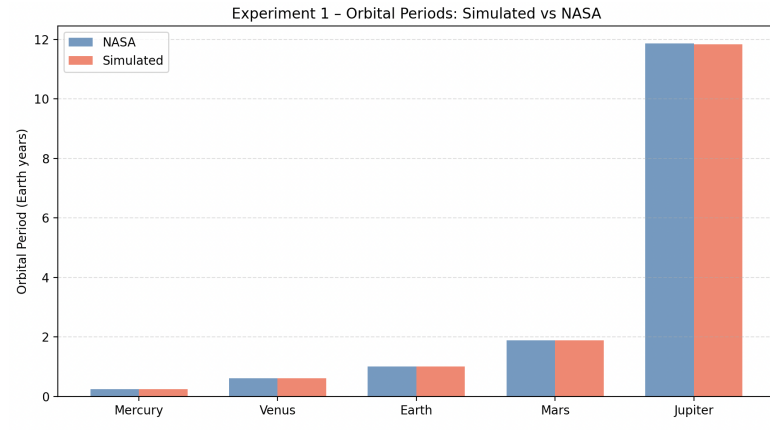


Figure 3. Bar charts comparing simulated data to NASA's data on orbital periods of planets.

2.5 Experiment 2

The first graph from this experiment, figure 4, depicts the fractional energy change, ΔE_{norm} , against time. There is a significant difference between the Beeman and Euler-Cromer integration methods and Direct Euler: the first two clearly obey the law of conservation of energy, whereas the latter very quickly diverges. Figure 5 shows the fractional energy change over time of just the Beeman method. Within less than 0.5

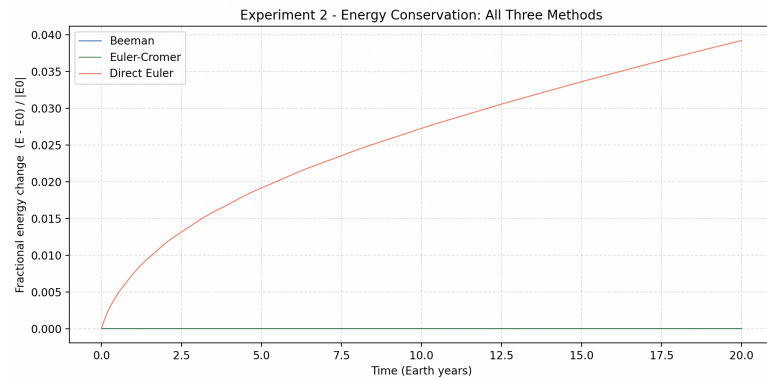


Figure 4. Fractional energy change against time for all three methods.

years, the change in energy plateaus. This is an ideal result as there are no visible perturbations after reaching a fractional difference of 4×10^{-6} , meaning that the method is stable and can be reliably used to model planetary motion over a large time scale.

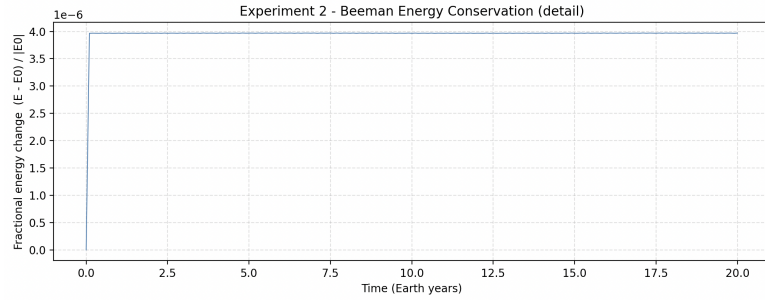


Figure 5. Fractional energy change of the Beeman simulation in greater detail.

2.6 Experiment 3

The simulation produced an elliptical transfer orbit connecting the orbit of Earth to that of Mars, as shown in figure 6. The probe follows an elliptical path consistent with what would be expected of a Hohmann transfer orbit², however, the relative position of Mars would need to be at a specific distance from Earth at launch in order for the probe to then orbit Mars. This was not incorporated into the model as it would be far more complex to code.

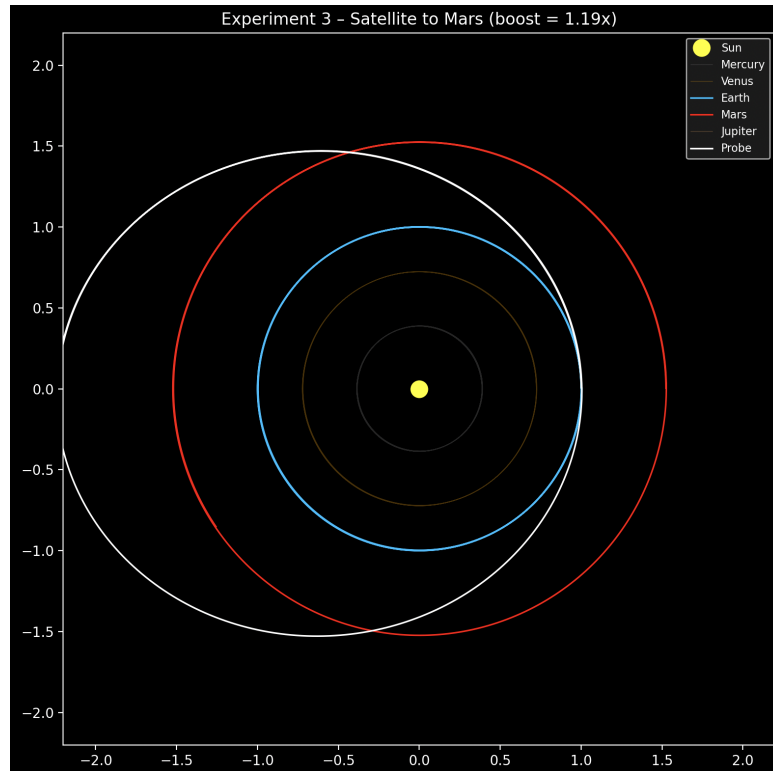


Figure 6. A view of the orbits of various planets and the probe at its optimal velocity.

3 DISCUSSION

In experiment 1, it is likely that the underestimate in orbital period is due to the assumptions made in the code, such as a perfectly circular orbit, constant speed, and no other bodies existing, whereas in the real world, there could be fluctuations in direction, speed, and objects in the surrounding area, as well as satellites around each body that could affect the acceleration due to gravity on that body.

As for experiment 2, the divergence of the Direct Euler line in figure 4 suggests that, as a simple integration method, Direct Euler could be used over very short time spans, but for any significant amount of time, the

Euler-Cromer or Beeman methods are needed.

NASA's Perseverance rover took 203 days - 0.556 years - to reach Mars⁵. However, the journey time in experiment 3 is unlikely to match exactly for several reasons: Perseverance used a specifically timed launch window where Earth and Mars were in ideal relative positions, whereas the simulation starts all planets on the same side of the Sun, which is not a realistic launch configuration; and Perseverance used trajectory correction manoeuvres which this simulation does not model.

4 CONCLUSION

The experiments performed all appear to be within alignment with real world data, suggesting that the code is suitable for modelling the solar system. It is clear that the accuracy would benefit from the code being altered to account for various starting alignments of planets but this was beyond the scope of the project.

5 BIBLIOGRAPHY

1. DuBois S. Hohmann Transfer [Internet]. Marspedia. 2024 [cited 2026 Apr 1]. Available from: https://marspedia.org/Hohmann_transfer
2. NASA. Trajectories - NASA Science [Internet]. science.nasa.gov. 2024 [cited 2026 Apr 1]. Available from: <https://science.nasa.gov/learn/basics-of-space-flight/chapter4-1/>
3. Amini M, Eastwood JW, Hockney RW. Time integration in particle models. Computer Physics Communications [Internet]. 1987;44:83–93. Available from: <https://www.sciencedirect.com/science/article/pii/0010465587900191>
4. NASA. Planetary Fact Sheet [Internet]. Nasa.gov. NASA; 2018 [cited 2026 Mar 20]. Available from: <https://nssdc.gsfc.nasa.gov/planetary/factsheet/>
5. NASA. Mars 2020: Perseverance Rover - NASA Science [Internet]. science.nasa.gov. NASA; 2025 [cited 2026 Mar 31]. Available from: <https://science.nasa.gov/mission/mars-2020-perseverance/>
6. HOHMANN W. THE ATTAINABILITY OF HEAVENLY BODIES [Internet]. Internet Archive. NASA; 1960 [cited 2026 Apr 2]. Available from: https://archive.org/details/nasa_techdoc_19980230631

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